Convergence Proofs of DP

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Some slides are from

10703 Deep Reinforcement Learning

Solving known MDPs

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September 10, 2018

Many slides borrowed from
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I. Is Iterative <u>Policy Evaluation</u> Guaranteed to Converge?

Contraction Mapping Theorem

Definition:

An operator F on a normed vector space $\mathcal{X}$ is a γ**-contraction**, for $0 < \gamma < 1$, provided for all $x, y \in \mathcal{X}$

$$||F(x) - F(y)|| \leq \gamma ||x - y||$$

Theorem (Contraction mapping)

For a γ-contraction F in a complete normed vector space $\mathcal{X}$

- Iterative application of F converges to a unique fixed point in $\mathcal{X}$ independent of the starting point
- at a linear convergence rate determined by γ

Is IPE a contraction mapping?

Value Function Space

- Consider the vector space V over value functions
- There are $|\mathcal{S}|$ dimensions
- Each point in this space fully specifies a value function $\mathrm{v}(s)$
- Bellman backup is a contraction operator that brings value functions closer in this space (we will prove this)
- And therefore the backup must converge to a unique solution

Value Function ∞-Norm

- We will measure distance between state-value functions u and v by the ∞-norm
- i.e. the largest difference between state values:

$$||\mathrm{u} - \mathrm{v}||_\infty = \max_{s \in \mathcal{S}} |\mathrm{u}(s) - \mathrm{v}(s)|$$

Bellman Expectation Backup is a Contraction

- Define the Bellman expectation backup operator

$$F^\pi(\mathrm{v}) = r^\pi + \gamma T^\pi \mathrm{v}$$

- This operator is a γ-contraction, i.e. it makes value functions closer by at least γ,

$$\begin{aligned}
||F^\pi(\mathrm{u}) - F^\pi(\mathrm{v})||_\infty &= ||(r^\pi + \gamma T^\pi \mathrm{u}) - (r^\pi + \gamma T^\pi \mathrm{v})||_\infty \\
&= ||\gamma T^\pi(\mathrm{u} - \mathrm{v})||_\infty \\
&\leq ||\gamma T^\pi(\mathbb{1} \cdot ||\mathrm{u} - \mathrm{v}||_\infty)||_\infty \\
&\cancel{\leq} ||\gamma(T^\pi \mathbb{1}) \cdot ||\mathrm{u} - \mathrm{v}||_\infty)||_\infty \\
&\cancel{\leq} \gamma ||\mathrm{u} - \mathrm{v}||_\infty
\end{aligned}$$

Note we define $\mathbb{1} \equiv [1,1,\ldots,1]^T$

Note $T^\pi \mathbb{1} = \mathbb{1}$

Convergence of Iterative Police Evaluation

- The Bellman expectation operator F^π has a unique fixed point
- $\mathbf{v}_\pi$ is a fixed point of F^π (by Bellman expectation equation)
- By contraction mapping theorem: Iterative policy evaluation converges on $\mathbf{v}_\pi$

II. Is Iterative Policy Iteration Guaranteed to Converge?

- Do policy iteration with greedy policy improvement converges to Q^* and π^* ?

Sketch

1. Show monotone improvement: $V^{\pi_{k+1}} \geq V^{\pi_k}$
2. Show that the algorithm have to stop after a finite number of steps
3. Show the converged policy is the optimal policy

1. Montone Policy Improvement

- Consider a deterministic policy, $a = \pi(s)$
- We can *improve* the policy by acting greedily

$$\pi'(s) = \operatorname*{argmax}_{a \in \mathcal{A}} q_\pi(s, a)$$

- This improves the value from any state s over one step,

$$q_\pi(s, \pi'(s)) = \max_{a \in \mathcal{A}} q_\pi(s, a) \geq q_\pi(s, \pi(s)) = v_\pi(s)$$

1. Montone Policy Improvement

- It therefore improves the value function, $v_{\pi'}(s) \geq v_\pi(s)$

$$
\begin{aligned}
v_\pi(s) &\leq q_\pi(s, \pi'(s)) = \mathbb{E}_{\pi'}\left[R_{t+1} + \gamma v_\pi(S_{t+1}) \mid S_t = s\right] \\
&\leq \mathbb{E}_{\pi'}\left[R_{t+1} + \gamma q_\pi(S_{t+1}, \pi'(S_{t+1})) \mid S_t = s\right] \\
&\leq \mathbb{E}_{\pi'}\left[R_{t+1} + \gamma R_{t+2} + \gamma^2 q_\pi(S_{t+2}, \pi'(S_{t+2})) \mid S_t = s\right] \\
&\leq \mathbb{E}_{\pi'}\left[R_{t+1} + \gamma R_{t+2} + \ldots \mid S_t = s\right] = v_{\pi'}(s)
\end{aligned}
$$

2. Convergence

- Since there exists only a finite number of policies, the algorithm has to stop after a finite number of steps

3. The Converged Policy is to the Optimal Policy

- If improvements stop,

$$q_\pi(s, \pi'(s)) = \max_{a \in \mathcal{A}} q_\pi(s, a) = q_\pi(s, \pi(s)) = v_\pi(s)$$

- Then the Bellman optimality equation has been satisfied

$$v_\pi(s) = \max_{a \in \mathcal{A}} q_\pi(s, a)$$

- Therefore $v_\pi(s) = v_*(s)$ for all $s \in \mathcal{S}$
- so π is an optimal policy

III. Is <u>Value Iteration</u> Guaranteed to Converge?

Bellman Optimality Backup is a Contraction

- Define the *Bellman optimality backup operator* T^*,

$$T^*(v) = \max_{a \in \mathcal{A}} \mathcal{R}^a + \gamma \mathcal{P}^a v$$

- This operator is a γ-contraction, i.e. it makes value functions closer by at least γ (similar to previous proof)

$$||T^*(u) - T^*(v)||_\infty \leq \gamma ||u - v||_\infty$$

Convergence of Value Iteration

- The Bellman optimality operator T^* has a unique fixed point
- v_* is a fixed point of T^* (by Bellman optimality equation)
- By contraction mapping theorem
- Value iteration converges on v_*